



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES
First Semester, Academic Year, 2026 – 2027



ITEP 414 - System Administration and Maintenance

Week 2 Portfolio Project: Enterprise Infrastructure Planning for a Startup Company

Bachelor of Science in Information Technology

Enterprise Infrastructure Plan

QWERTY Solution Inc.

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Part 1- Company Profile

Company Name: QWERTY Solution Inc.

Nature of Business: A software development and IT consulting startup building custom web and mobile applications, and providing digital-transformation services for small and medium-sized enterprises (SMEs) in the Philippines.

Vision: To become a leading Philippine-based software solutions provider recognized across Southeast Asia for innovation, reliability, and client-centered technology.

Mission: To deliver high-quality, scalable, and secure software solutions; to nurture a culture of continuous learning and technical excellence among our people; and to help our clients grow through well-engineered technology.

Office Location (fictional): Brgy. Palasan Santa Cruz Laguna, Philippines

Organizational Structure

The company follows a flat, functional structure appropriate for a 20-person startup:

- Chief Executive Officer (CEO)
- IT Manager - oversees the IT Department (5 staff, incl. the Junior System Administrator)
- HR Manager - oversees the HR Department (4 staff)
- Finance Manager - oversees the Finance Department (5 staff)
- Sales Manager - oversees the Sales Department (6 staff)



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- Employee Distribution

Department	Employees
Information Technology	5
Human Resources	4
Finance	5
Sales	6
TOTAL	20

Part 2 - Enterprise Hardware Inventory

Quantities are matched 1:1 to headcount for primary compute devices (14 desktops + 6 laptops = 20 employees), with shared infrastructure sized for the whole office.

Asset ID	Hardware	Qty	Department	Purpose
HW-001	Desktop Computer	5	IT Department	Dev workstations for coding, testing, and running local VirtualBox environments.
HW-002	Desktop Computer	4	HR Department	Standard desktops for records management, recruitment, and payroll support.
HW-003	Desktop Computer	5	Finance Department	Standard desktops for accounting, invoicing, and financial reporting.



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HW-004	Laptop	6	Sales Department	Portable units for client visits, demos, and field/remote sales work.
HW-005	Server (Rack-mount)	1	Server Room (IT)	Hosts file sharing, internal Git repository, and internal test apps (Ubuntu Server).
HW-006	Router	1	Network Infra.	Gateway routing, NAT, and DHCP between the LAN and the ISP connection.
HW-007	Switch (24-port Managed)	2	Network Infra.	Wired connectivity to all departments with VLAN segmentation.
HW-008	Network Printer	4	All Departments	One shared, network-attached printer per department to control cost.
HW-009	UPS	6	Server Room & Depts.	Power protection for the server and critical workstation clusters.
HW-010	Wireless Access Point	3	Office-wide	Wi-Fi coverage across the floor for laptops and mobile devices.
HW-011	NAS Storage	1	Server Room (IT)	Centralized RAID-protected storage for shared files and nightly backups.
HW-012	External Backup Drive	2	IT Department	Rotated offsite backup copies for disaster recovery.
HW-013	Monitor	16	IT, HR, Finance, Meeting Rms.	One per desktop user (14) plus 2 for meeting/conference rooms.



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Part 3 - Enterprise Software Inventory

All licenses are procured under per-device or per-user volume agreements where applicable, and tracked in the IT asset register to remain compliant during audits.

Software	Version	License	Purpose
Windows 11 Pro	23H2	Volume License (per device)	Primary endpoint OS; supports domain join, BitLocker, and centralized policy management.
Ubuntu Server	22.04 LTS	Free / Open Source	OS for the internal server: file sharing, Git hosting, and internal test apps.
Microsoft 365 / Office	Current (2024 Apps)	Subscription, per user	Word, Excel, Outlook, Teams for documentation, reporting, and communication.
VS Code	Latest stable	Free / Open Source	Primary IDE for the IT/development team.
Git	Latest stable	Free / Open Source	Version control for all software projects.
GitHub Desktop	Latest stable	Free / Open Source	Simplified Git GUI client for non-CLI-comfortable staff.
VirtualBox	Latest stable	Free / Open Source (GPL)	Local virtual machines for testing, sandboxing, and training.



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Google Chrome	Latest stable	Free	Standard web browser for daily operations.
Microsoft Defender	Included w/ Win 11	Included / Free	Built-in endpoint antivirus and threat protection, centrally managed.
AnyDesk	Latest stable	Free / Business tier	Remote access for IT helpdesk support and troubleshooting.
7-Zip	Latest stable	Free / Open Source	File compression and archiving utility.

Part 4 - Enterprise Network Inventory

Asset ID	Equipment	Qty	Specification	Purpose
NW- 001	ISP Modem	1	Fiber modem, provided by ISP	Converts the incoming ISP fiber line into a usable network handoff.
NW- 002	Router	1	Enterprise Gigabit router	Routes traffic between LAN and internet; handles NAT and DHCP.
NW- 003	Firewall	1	Hardware UTM appliance	Perimeter security: intrusion prevention, content filtering, VPN.
NW- 004	Switch	2	24-port Gigabit managed	Distributes wired connectivity; VLAN segmentation per department.



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NW-005	Access Point	3	Dual-band Wi-Fi 6	Wireless coverage for laptops and mobile/guest devices.
NW-006	Patch Panel	1	24-port Cat6	Organizes and terminates structured cabling runs.
NW-007	CAT6 Cables	1 box (305m) + patch cables	Category 6, UTP	Structured cabling for all wired workstation and device drops.
NW-008	RJ45 Connectors	100 pcs	Cat6-rated	Cable termination for on-site crimping, moves, adds, and repairs.



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Part 5 - Enterprise Network Diagram

The diagram below shows the logical topology: internet access flows through the ISP modem, firewall, and router before reaching the core switch, which distributes connectivity to the server room, shared services, wireless access points, and each department's LAN segment.

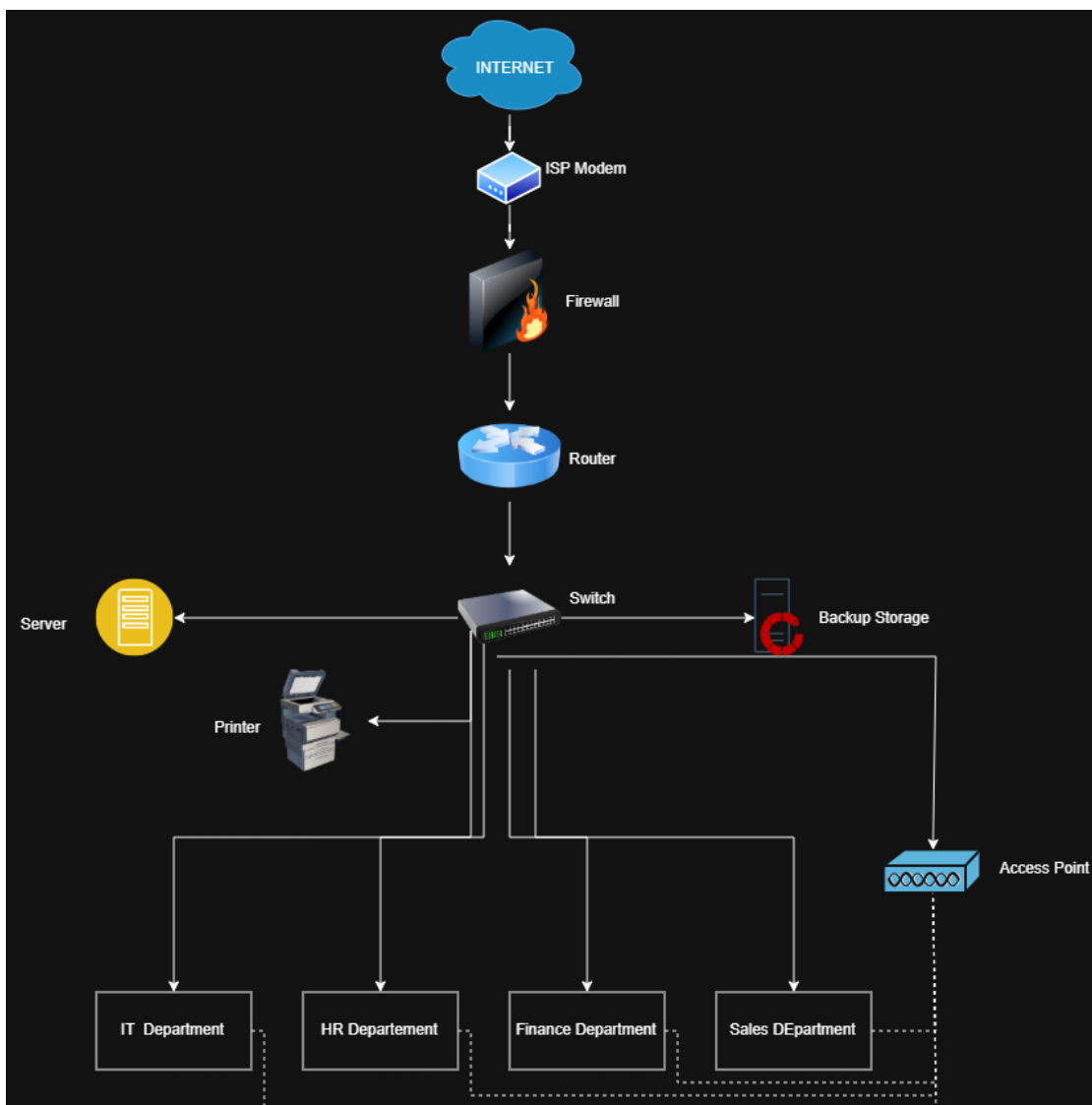


Figure 1. QWERTY Solutions Inc. - Enterprise Network Topology



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Part 6 - System Administration Roles

Helpdesk Technician

Responsibilities: Provides first-line technical support to employees resolving tickets, troubleshooting hardware/software issues, resetting passwords and accounts, and setting up new-hire workstations.

Key Skills: Customer service, basic Windows/Office troubleshooting, clear communication, patience under pressure, ticket prioritization.

Common Tools: Remote support software (AnyDesk, TeamViewer), helpdesk ticketing systems (Freshservice, Zendesk), Active Directory basics.

Relevant Certifications: CompTIA A+, ITIL Foundation, Microsoft 365 Certified: Fundamentals.

Network Administrator

Responsibilities: Configures and maintains routers, switches, firewalls, and wireless access points; monitors network performance and uptime; manages VLANs and network security policies.

Key Skills: TCP/IP, DNS, DHCP, routing and switching, firewall configuration, network monitoring and troubleshooting.

Common Tools: Vendor CLI (Cisco/MikroTik), Wireshark, PRTG or Nagios for monitoring, pfSense/UTM consoles.

Relevant Certifications: CCNA, CompTIA Network+, Fortinet NSE.



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Linux System Administrator

Responsibilities: Manages the internal Ubuntu Server user accounts and permissions, package updates, backup jobs, automation scripts, and general server uptime and security hardening.

Key Skills: Linux command line, Bash scripting, package management (apt), cron scheduling, basic firewall/security hardening.

Common Tools: SSH, Bash, systemd, cron, rsync, UFW/iptables.

Relevant Certifications: LPIC-1, RHCSA Red Hat Certified System Administrator, CompTIA Linux+.

Cloud Administrator

Responsibilities: Manages cloud-hosted resources such as offsite backup storage, evaluates cloud migration options, monitors cloud spend, and plans for scaling infrastructure as the company grows.

Key Skills: Cloud fundamentals (AWS/Azure/GCP), identity and access management, virtualization, basic cost optimization and automation.

Common Tools: Cloud provider consoles (AWS/Azure/GCP), Terraform (infrastructure-as-code), CLI tools.

Relevant Certifications: AWS Certified Cloud Practitioner, Microsoft Azure Fundamentals -AZ-900.



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How These Roles Work Together

In a 20-person startup like QWERTY Solutions, these four roles form a single, interdependent support chain rather than isolated silos. The Helpdesk Technician is the first point of contact for everyday issues and escalates anything beyond basic troubleshooting. The Network Administrator keeps the underlying connectivity routing, switching, Wi-Fi, and firewall rules healthy and secure, which every other role depends on to do their job. The Linux System Administrator keeps the internal server, file sharing, and Git hosting available and backed up, directly supporting the developers and the company's day-to-day file access. The Cloud Administrator extends that same responsibility outward, managing offsite backups today and planning the company's eventual move to cloud-hosted infrastructure as headcount grows. Together, they cover the full stack from the end-user's desk to the internet edge to offsite storage, and they coordinate closely for example, a Helpdesk ticket about a slow shared drive might involve the Network Administrator checking bandwidth, the Linux Administrator checking server load, and the Cloud Administrator checking whether an offload to cloud storage is warranted.

Part 7 - Infrastructure Recommendations

Internet Provider: A business-grade fiber connection (an enterprise plan from PLDT Enterprise or Converge ICT Business) with a service-level agreement (SLA) for guaranteed uptime and symmetric upload/download speeds important for Git pushes, client demos, and cloud backups. A secondary ISP or LTE failover line is recommended for redundancy.



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Server Specifications: A rack-mount server with a modern server-grade CPU (Intel Xeon E-2400 series or AMD EPYC entry-level), 32–64 GB ECC RAM, RAID 1 or RAID 5 SSD storage for resilience, dual (redundant) power supplies, and Ubuntu Server 22.04 LTS as the OS.

Backup Strategy: Follow the 3-2-1 rule recommended by CISA for small and medium businesses at least 3 copies of data, on 2 different media types, with 1 copy offsite. In practice: primary data on the server, a nightly automated backup to the NAS, and a rotated external drive (or cloud backup) kept offsite. Backups should be tested quarterly to confirm they can actually be restored.

Security Recommendations: Segment the network with VLANs per department, enforce the firewall as the single perimeter chokepoint, keep all systems patched on a regular schedule, and apply the principle of least privilege so staff only access what their role requires.

Antivirus: Microsoft Defender for Business (or Endpoint), centrally managed through a dashboard so the IT team can monitor threats and push policy across every device, rather than relying on unmanaged built-in protection alone.

Password Policy: Following NIST's digital identity guidelines, prioritize password length over forced complexity a minimum 12-character passphrase, mandatory multi-factor authentication (MFA) for email and remote access, no shared or generic accounts, and no forced periodic rotation (which encourages weak, predictable changes). Use a company password manager to avoid weak or reused passwords.

Expansion Plan: The design leaves headroom for growth: the core switch has spare ports, the wireless APs are positioned for a larger floor if needed, and the backup/cloud strategy is already



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cloud-ready. As the company scales past roughly 50 employees, QWERTY should evaluate a hybrid-cloud model keeping sensitive data on-premises while moving compute-heavy or public-facing workloads to a cloud provider.

Part 8 - Personal Reflection

Working on this Week 2 portfolio project helped me understand more clearly what a System Administrator does when planning an IT infrastructure. At first, I thought infrastructure planning was mainly about choosing computers, servers, and network devices. However, while working on the project, I realized that planning starts with understanding the needs of the business and its employees. I had to consider the number of employees in each department, the type of work they perform, and the hardware, software, and network resources they would need. This helped me understand that every IT equipment and software choice should have a specific purpose.

One of the most challenging parts of the project for me was creating the enterprise network diagram. Listing the network equipment was easier, but connecting the modem, firewall, router, switches, server, access points, and different departments required more careful planning. I had to think about how information would travel through the network and how the different devices would work together. This part helped me improve my understanding of network structure and made me realize how important organization and proper design are when building an enterprise network.

I also learned that security and backup planning are important parts of system administration. The project included VLAN segmentation, firewall protection, user permissions, strong passwords, multi-



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factor authentication, and the 3-2-1 backup strategy. These requirements showed me that maintaining an IT infrastructure is not only about making systems work but also about protecting company data and making sure services can continue even when problems occur.

Overall, this project gave me a better understanding of how hardware, software, networking, security, and system administration are connected. It also helped me practice making decisions based on the actual needs of a company instead of simply choosing equipment without a reason. I feel that this activity improved my confidence and gave me a better idea of what I may be expected to do as a future IT professional or System Administrator. I also learned that good planning can prevent unnecessary expenses, technical problems, and security risks in the future.



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